

Matched Outcomes, Divergent Gaze: How Foveated MLLMs Search Compared to Humans *

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Abstract. Human visual search is *serial*: the fovea must land on a candidate to confirm it, producing a scanpath. Whether multimodal large language models (MLLMs), constrained to the same foveated input, search as humans do bears on their use as models of human vision and on attention-alignment scores. Three general-purpose MLLMs are compared with human eye-movement scanpaths on goal-directed search (COCO-Search18), each driven fixation-by-fixation through an identical human-matched foveated view and assessed along three axes: the *decision* (target present or absent), the *efficiency* of reaching the target, and the *gaze process* itself. The axes dissociate. On the decision and on target acquisition the models match or exceed humans, detecting present targets near ceiling and reaching them on the first saccade more often than humans. The gaze process is not human: all three share one signature—low-entropy, large-amplitude, self-consistent scanpaths that agree with themselves far more closely than two humans agree. This is consistent with a single-pass, non-serial architecture rather than a limit of acuity: matched retinal input reproduces where humans look but not how looking unfolds, and no degree of peripheral degradation recovers human-like search. The difference lies on a process axis that answer-alignment and saliency metrics do not measure; such metrics cannot certify human-like vision, and zero-shot models suit outcome and spatial questions, not temporal, process-level ones.

Keywords: Visual search · Foveated vision · Multimodal LLMs · Eye movements · COCO-Search18 · Human-model alignment

1 Introduction

Human goal-directed visual search is a *serial* process. The fovea resolves fine detail only within $\sim 1\text{--}2^\circ$ of gaze, so a target detected coarsely in the periphery must be fixated before it can be confirmed; search unfolds as a sequence of saccades whose targeting, extent and termination are well characterised, theoretically [16, 17], computationally [4, 33, 38, 39], and empirically on datasets such as COCO-Search18 for target-present and target-absent search [6, 7]. This seriality

* The recorded data and the analysis code required to reproduce every result will be released upon publication.

is imposed by the optics of the eye and is what a foveated agent must pay to search.

This motivates a falsifiable hypothesis: if a model views a scene through the *same* foveated, acuity-limited aperture as a human—sharp at gaze, degraded in the periphery, displaceable only by re-fixating—then matched retinal input ought to induce matched search behaviour. The assumption is load-bearing for two common practices: using multimodal large language models (MLLMs) as stand-ins for human observers, and reading attention-alignment scores (the overlap between model attention and human fixations) as evidence that a model “sees like us”. Both take behavioural or spatial agreement to license a claim about *process*. We test this by decomposing “human-like search” into three axes and asking on which, if any, an MLLM resembles a human: *decision* (target present or absent?), *finding* (does gaze reach the target, at what cost?), and *gaze* (are the eye-movement dynamics human-like?). Three general-purpose MLLMs from three families (Qwen3.5-35B-A3B [28], GLM-4.6V-Flash [34], Gemma-4-E4B [14]) search COCO-Search18 fixation-by-fixation through a byte-identical foveation harness, against ten human scanpaths per scene.

The axes *dissociate* (Fig. 1). On decision and finding the models are human-or-better—near-ceiling present-target detection and first-saccade target fixation 0.97/0.97/0.80 versus the human 0.49, at comparable eventual success and no more fixations—yet on gaze they are non-human, the three sharing *one* low-entropy, large-amplitude, highly self-consistent signature that agrees with itself far more than with any human (cross-seed ScanMatch 0.84/0.91/0.71 against the inter-observer ceiling 0.53). The correct outcome is produced by a shared, non-human process—consistent with a single-pass, non-serial architecture rather than a limit of acuity, as matched retinal *input* reproduces *where* humans look but not *how* looking unfolds (§4.2). Our contributions follow: answer-alignment and saliency metrics are blind to this process axis and cannot certify human-like vision; zero-shot MLLMs suit outcome and spatial questions but not process and temporal ones; and a non-serial searcher that matches human outcomes is a null model for the human serial bottleneck. We do not propose a scanpath predictor or compete on accuracy.

2 Related work

Human search and its models. COCO-Search18 provides laboratory human fixations for target-present and target-absent search [6, 7] and anchors scanpath predictors—inverse-reinforcement-learning and transformer models [27, 38, 39] and ideal-observer/Bayesian searchers benchmarked on common data [4, 33]. Classically, search mixes parallel peripheral evaluation with serial focal inspection [17], and search asymmetries emerge from natural-image statistics rather than task-specific training [16]. These *predict or explain human* attention; we characterise an MLLM’s *process* against the same reference, taking seriality as the property we test.

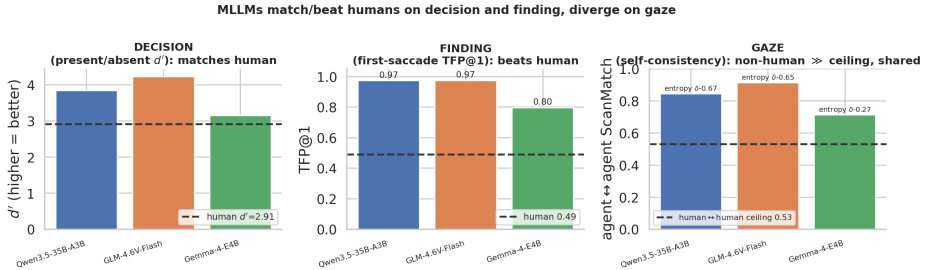


Fig. 1: The three-axis dissociation at the human-matched condition. *Decision* (d' , present/absent): the models match or exceed the human reference. *Finding* (first-saccade TFP@1): the models exceed the human rate (0.49) by a wide margin. *Gaze* (cross-seed self-consistency): all three lie above the human $\leftrightarrow$ human ceiling (0.53) and apart from the human (bar height, cross-seed ScanMatch; annotation, gaze-entropy Cliff’s δ), with Gemma-4-E4B nearest. The correct outcome is produced by a shared, non-human process.

Foveation as a modelling constraint. The Geisler–Perry acuity fall-off [13] supplies our human-matched renderer. Foveated-observer models capture human performance that non-foveated metrics miss—medical search [20], foveated transformers [18], scene-understanding time [35], dual “what”/“where” saccade selection [9]—and semantically guided foveal models predict human scanpaths on COCO-Search18 [24,25]. Most pointedly, biologically constrained networks viewing scenes foveally produce human-like scanpaths *without being trained to* [29,41]. That foveation has repeatedly *sufficed* to induce human-like search motivates our question; for a general-purpose MLLM it does not (Sec. 4.3).

MLLM versus human vision. A growing literature asks whether MLLMs perceive as humans do—cognitive paradigms [5,10,12,23], which find that models “see but do not perceive”, and attention comparison via eye-tracking [15,31,32] where attention similarity and task performance dissociate [32]; critiques of linguistic-prior reliance separate spatial from semantic guidance [19,36]. Closest, MLLM competence and the human process come apart: a serial deficit inferred from reaction time [3], human-trivial search on which VLMs barely beat chance [2], correct region with wrong answer [42]—the mirror of our right-answer/non-human-process finding, but on a static map—and a capability–strategy gap accuracy benchmarks miss [26]. We differ in measuring the fixation-by-fixation process, with an explicit stopping decision, on an axis these do not assess.

Agentic, search-trained MLLMs. A parallel line adds active perception to raise accuracy—LLM-guided search [37], tree-based zoom [30], reinforcement-learned focusing [1,21,22], and embodied variants [11,40]; chain-of-thought can even *degrade* an embodied searcher [43]. These optimise *what* is answered by learning *where* to look. We instead characterise *how* a general-purpose model explores under a fixed foveation constraint, without search-specific training; trained agents are the most pertinent next comparison (Sec. 6) but out of scope.

3 Method

Data. The human reference is COCO-Search18 [6]: ten observers per scene issuing a gamepad present/absent decision on 1680×1050 px scenes subtending $\sim 54^\circ \times 35^\circ$ ($\rightarrow \sim 30$ px/deg). We use the validation split and a frozen, category-stratified subset of 141 target-present and 144 target-absent scenes (all ten human scanpaths each); the unit of analysis is the (scene, target) trial. **Foveation bracket.** At the gaze point a deterministic renderer applies one of four condition *families* (nine conditions in total; Fig. 2): SHARP; GEISLER-PERRY (GP), the Geisler and Perry [13] acuity falloff (the only human-matched condition; its mild appearance reflects that peripheral information at this viewing geometry is more legible than intuition suggests, and follows the calibrated falloff of Sec. S1); GAUSSIAN “gist- k ” (“gist” denotes the coarse peripheral information surviving foveation), the same falloff with peripheral acuity reduced by $k \in \{8, 16, 24, 32, 48, 128\}$ (a synthetic degradation not tuned to human behaviour); and CROP, a fovea-only disc. **Procedure.** Each episode begins at a forced central fixation; at every step the model observes the scene rendered at its gaze point (earlier glimpses retained in context) and returns one directive: LOOK, FOUND, or ABSENT. We use “scanpath”/“gaze” for the model’s sequence of requested fixation coordinates: an operational analogue of oculomotor scanning, not a claim that the model has eye movements. No search policy is imposed; the 50-glimpse cap never forces termination (episodes end on the model’s own FOUND/ABSENT decision). Decisions use a single free-form generation per step at temperature 0.6 under 5 seeds; the harness, prompt (see the supplementary material) and renderer are identical across models. The Geisler-Perry falloff is computed in the native 1680×1050 frame and each glimpse is then downsampled uniformly to a 1024-px maximum side, preserving the falloff geometry up to a global scale. **Models.** Qwen3.5-35B-A3B (a 35B mixture-of-experts, 3B active; reasoning-tuned), GLM-4.6V-Flash (≈ 9 B; reasoning-tuned) and Gemma-4-E4B (≈ 4 B; instruction-tuned, run with thinking disabled), which differ in architecture, family and training recipe; as these factors covary, any cross-model trend is reported descriptively.¹ **Measures.** Search is interpreted only on trials passing the one-shot existence test, whose detection ceiling is near-identical across models (Sec. 4.1). We report, for *decision*, existence accuracy and signal-detection d' (criterion in Sec. S4); for *finding*, target-fixation probability (TFP) by saccade, with a hit defined as a fixation within 1° of the target box, and fixation count; and for *gaze*, the per-scanpath signature (saccade amplitude, entropy, refixation, length, center bias, turning angle, coverage) as Cliff’s δ against humans, scanpath similarity [8] against the human $\leftrightarrow$ human ceiling, and a cross-model PCA. All measures are defined in Sec. S3; durations are human-only.

¹ HuggingFace checkpoints Qwen/Qwen3.5-35B-A3B, zai-org/GLM-4.6V-Flash, google/gemma-4-E4B-it. The two reasoning-tuned models run with their default thinking and Gemma-4-E4B with thinking disabled; at each step only the final directive line is consumed.



Fig. 2: The foveation bracket applied to one scene at a fixed gaze point (red +). Only GEISLER-PERRY is human-matched, and at this geometry it is mild, nearly indistinguishable from SHARP; the gaussian gist ladder and CROP are synthetic anchors.

Table 1: Outcome by foveation condition (existence-passed trials); each entry is Qwen/GLM/Gemma (Q/G/Gm) against the human reference (first row; per-model n differs slightly). Finding and outcome measures match or exceed the human values; the *dynamic* gaze divergence is quantified in Table 2, though target-absent search length (NFix-TA) also differs from humans for Qwen. TFP@1/TFP-end: target fixated on the first / by the final saccade (TP); NFix-TP/-TA: median fixations on target-present/absent trials; TA decl.-abs: target-absent declared-absent rate; dens. CC: correlation of model and human fixation-density maps.

condition	TFP@1			TFP-end			NFix-TP			NFix-TA			TA decl.-abs			dens. CC		
	Q	G	Gm	Q	G	Gm	Q	G	Gm	Q	G	Gm	Q	G	Gm	Q	G	Gm
human	0.49			0.93			3			5			—			—		
sharp	0.97	0.97	0.80	0.97	0.98	0.93	2	2	3	1	4	6	0.94	0.91	0.96	0.55	0.54	0.49
GP	0.97	0.97	0.80	0.98	0.98	0.93	2	2	3	1	4	5	0.93	0.91	0.96	0.58	0.63	0.50
k8	0.93	0.84	0.64	0.95	0.89	0.86	3	2	3	2	4	6	0.84	0.77	0.81	0.65	0.45	0.26
k16	0.78	0.71	0.42	0.91	0.80	0.75	3	3	4	4	4	6	0.78	0.77	0.68	0.70	0.38	0.45
k24	0.56	0.49	0.28	0.81	0.65	0.65	3	3	6	6	4	6	0.72	0.68	0.58	0.74	0.46	0.29
k32	0.42	0.33	0.19	0.71	0.53	0.59	3	3	6	7	4.5	7	0.62	0.65	0.52	0.70	0.43	0.26
k48	0.28	0.18	0.10	0.56	0.36	0.47	5	4	6	7	4	7	0.46	0.65	0.54	0.49	0.33	0.25
k128	0.15	0.09	0.01	0.31	0.22	0.39	6	4	6	6	4	6	0.85	0.86	0.78	0.18	0.14	0.19
crop	0.04	0.04	0.00	0.20	0.11	0.16	3	3	5	2	3	5	0.68	0.50	0.67	0.25	0.11	0.15

4 Results

The three models were evaluated on 285 scenes, 5 seeds and nine conditions (3 models $\times$ 285 scenes $\times$ 5 seeds $\times$ nine conditions = 38,475 episodes), all completed (Sec. S10). Quantities are reported at the human-matched GP condition unless a sweep is specified; complete per-condition tables for all three models appear in the supplement.

4.1 Decision and finding: models match or exceed humans

Decision. One-shot present-target detection is near-ceiling and comparable across models (target-present / target-absent existence accuracy 0.99/0.81, 0.99/0.81, 0.97/0.80; false-positive bias on absent scenes 0.19/0.19/0.20), so subsequent

Table 2: Gaze axis at the human-matched condition: Cliff’s δ against the human distribution ($|\delta| > 0.33$ in bold; sign is agent – human) and cross-seed self-consistency. The deviation is non-human and shares a common direction across models (sub-threshold for Gemma-4-E4B on entropy and amplitude), whereas center bias, the spatial prior, is human-like (sub-threshold) for all three. Full seven-statistic Cliff’s δ (all conditions): Table S3.

	gaze entropy δ	saccade amp δ	center bias δ	agent $\leftrightarrow$ agent self-consist.
human $\leftrightarrow$ human ceiling	—	—	—	0.53
Qwen3.5-35B-A3B	−.67	+ .50	−.18	0.84
GLM-4.6V-Flash	−.65	+ .61	−.31	0.91
Gemma-4-E4B	−.27	+ .23	+ .06	0.71

search divergences cannot be attributed to detection failure and the existence-passed sets are comparable. In the agentic task the present/absent decision is highly sensitive for every model (d' 3.84/4.23/3.14, against the human 2.91; Sec. S4), and absent scenes are declared absent at 0.93/0.91/0.96 — distinct from the one-shot existence accuracy (~ 0.80) reported above. **Finding.** Every model fixates the target on the first saccade far more often than humans (TFP@1 0.97/0.97/0.80 against 0.49; Table 1) and attains comparable eventual success (TFP-end 0.98/0.98/0.93 against 0.93), while issuing no more fixations than humans (median 2/2/3 against 3). On the two axes of principal practical interest, whether the decision is correct and whether the target is found, the models are therefore human-or-better; assessed at the level of outcomes alone, they would be judged human-like. One target-absent behaviour is, however, already non-human on the search axis: Qwen3.5-35B-A3B declares absence after a single fixation (NFix-TA 1) versus the human median 5, an efficiency win that is itself not human-like.

4.2 Gaze dynamics: a shared, non-human signature

Under the conditions in which outcomes coincide with humans, the eye-movement *process* does not, and the deviation has the same direction for all three models (Table 2, Fig. 3b). Gaze entropy lies below the human value (Cliff’s δ −.67/−.65/−.27) indicating spatially concentrated sampling, and saccade amplitudes exceed it (+.50/+ .61/+ .23), indicating direct movements to the target. The scanpaths are moreover highly self-consistent: cross-seed (agent $\leftrightarrow$ agent) ScanMatch is 0.84/0.91/0.71, far above the human $\leftrightarrow$ human agreement ceiling (0.53), so each model reproduces its own scanpath more closely than two humans agree. A principal-component analysis of the five-statistic signature places all three models, under the legible conditions, in a region disjoint from the human reference (Fig. 3b). That three unrelated models occupy the *same* region is the multivariate expression of the gaze divergence. These effects survive per-metric mixed-effects models with crossed scene and rater random intercepts (gaze entropy and saccade amplitude have 95% intervals excluding zero for all three models; Sec. S9).

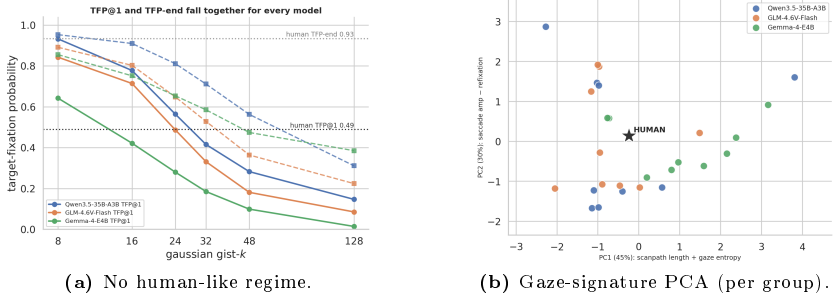


Fig. 3: (a) First-saccade targeting (solid) and eventual success (dashed) as functions of the degradation factor k for the three models, with human reference levels; the two quantities fall together, so no k yields human-like search at human-like success. (b) Principal-component analysis of the per-(model, condition) five-statistic gaze signature (PC1, 45%: scanpath length and gaze entropy; PC2, 30%: saccade amplitude versus refixation). Under the legible conditions each model sits in a region offset from the human reference ($\star$); points approach the human only under severe degradation, with Gemma-4-E4B nearest.

4.3 No regime recovers human-like search; a single-pass account

Sweeping the synthetic degradation does not yield a level at which a model searches like a human ($\text{TFP}@1 \approx 0.49$) while still locating the target ($\text{TFP-end} \approx 0.93$). For none of the three models (Fig. 3a) do these quantities separate: they decline together, so by the degradation that lowers first-saccade targeting to the human rate ($k=32$ for Qwen3.5-35B-A3B, $\text{TFP}@1/\text{TFP-end}$ 0.42/0.71; $k=16$ for Gemma-4-E4B, 0.42/0.75), eventual success has already collapsed. The divergence is not an effect of acuity: the human-matched condition is behaviourally indistinguishable from SHARP for every model (e.g. $\text{TFP}@1$ 0.97 against 0.97; Tables 1–2), even though it measurably degrades the periphery. Nor is it a difference of spatial prior: center bias is human-like for all three (sub-threshold δ , Table 2), so the models look in broadly human-relevant places, whereas the order and dynamics of their fixations (low entropy, large saccades, high determinism) do not. The missing component is serial sampling, not the spatial prior; we do not manipulate architecture directly, so this attribution is by elimination (acuity and spatial prior ruled out) and a causal test imposing serial sampling is left to future work. As evidence is removed the models do not lengthen search gracefully: refixation rises, then at the most severe degradation the declared-absent rate rebounds as they default to absent, a model-internal failure signature needing no human reference (Sec. S8).

4.4 Generality and robustness

The dissociation and the shared gaze signature hold across three distinct model families, suggesting a property of the foveated-MLLM paradigm rather than of a single instance. The first-saccade advantage holds within every eccentricity-by-size difficulty stratum (Sec. S9) and is insensitive to the target-box tolerance: across tolerances of 0.5, 1 and 1.5°, $\text{TFP}@1$ shifts by at most ~ 0.1 . The

high cross-seed determinism is not an artifact of low-temperature decoding; in a temperature sweep on our anchor model (Qwen3.5-35B-A3B) it remains well above the inter-observer ceiling even at temperature 1.0 on the legible conditions (Sec. S9). Cross-seed and human inter-observer agreement are not identical constructs (no matched human intra-observer baseline exists in COCO-Search18), so we treat the determinism gap as suggestive, resting it on this temperature-1.0 persistence. The *magnitude* of the gaze deviation follows a consistent ordering across models: Gemma-4-E4B is the most human-like model, with the smallest effect on five of the seven statistics (entropy, saccade amplitude, scanpath length, center bias, coverage; e.g. gaze-entropy δ $-.27$ vs. Qwen $-.67$; cross-seed Scan-Match 0.71 vs. 0.84), and its number of fixations (search extent) is the only fixed effect whose mixed-effects 95% interval contains zero (Sec. S9). Because architecture, recipe and sparsity covary across three models, we read this as weaker single-pass targeting, not a more human-like strategy, and report it descriptively.

5 Discussion

Metrics and surrogacy. The models match the present/absent answer and partially match the human saliency map (density correlation 0.58/0.63/0.50) yet diverge on every temporal measure of gaze. Because answer-alignment and saliency scores are computed on outcomes or a time-collapsed map, a high value is necessary but not sufficient: zero-shot MLLMs are adequate surrogates for *outcome and spatial* questions but not *process and temporal* ones, a property of the class that model selection does not remove.

A non-human searcher as a null model. Conversely, a system attaining human-or-better outcomes *without* a serial bottleneck is a useful null: it isolates the behaviours due to seriality from the detection and spatial priors it already reproduces.

6 Limitations and scope

Our scope is general-purpose models under a fixed foveation constraint: search-trained agentic, pointing-native and frontier closed-source models—the most pertinent next comparison—are not evaluated. The cross-model trend is confounded (architecture, recipe, sparsity) and reported descriptively; the gaze signature is measured under a single prompt whose memory clause may partly shape refixation; and fixation durations are human-only.

7 Conclusion

Under human-matched foveation, three MLLMs match or beat humans on decision and finding yet share *one* non-human gaze signature that no degradation regime makes human-like, the mark of a single-pass architecture with a human-like spatial prior. This *process* axis, which answer-alignment and saliency metrics miss, bounds MLLMs as human-vision surrogates and yields a null model for the serial bottleneck.

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